

Solutions to Exercises 1

Part A - Intro to Logic

Exercise 1 (Sheffer Stroke). Any logical constant and connector can be simulated with Sheffer strokes. In fact:

$$\begin{aligned}
 \neg\varphi &\equiv \varphi \uparrow \varphi \\
 1 &\equiv \varphi \uparrow \neg\varphi \equiv \varphi \uparrow (\varphi \uparrow \varphi) \\
 0 &\equiv \neg 1 \equiv (\varphi \uparrow (\varphi \uparrow \varphi)) \uparrow (\varphi \uparrow (\varphi \uparrow \varphi)) \\
 \varphi_0 \wedge \varphi_1 &\equiv \neg(\varphi_0 \uparrow \varphi_1) \equiv (\varphi_0 \uparrow \varphi_1) \uparrow (\varphi_0 \uparrow \varphi_1) \\
 \varphi_0 \vee \varphi_1 &\equiv (\neg\varphi_0 \uparrow \neg\varphi_1) \equiv ((\varphi_0 \uparrow \varphi_0) \uparrow (\varphi_1 \uparrow \varphi_1)) \\
 \varphi_0 \rightarrow \varphi_1 &\equiv \neg\varphi_0 \vee \varphi_1 \equiv (((\varphi_0 \uparrow \varphi_0) \uparrow (\varphi_0 \uparrow \varphi_0)) \uparrow (\varphi_1 \uparrow \varphi_1))
 \end{aligned}$$

Exercise 2 (Decision Procedure for SAT). First we rewrite the formulas using only the allowed symbols:

1. $((a \rightarrow b) \rightarrow a) \rightarrow a \equiv \neg(\neg(\neg a \vee b) \vee a) \vee a$
2. $\neg(a \wedge b) \rightarrow (\neg a \vee \neg b) \equiv \neg\neg(a \wedge b) \vee (\neg a \vee \neg b)$
3. $((\neg a \rightarrow b) \wedge (a \rightarrow b)) \rightarrow b \equiv \neg((\neg\neg a \vee b) \wedge (\neg a \vee b)) \vee b$

Then we derive $\neg F \vdash 0$ (we collapse sequences of SIMP):

1.

$$\frac{\frac{\neg(\neg(\neg(\neg a \vee b) \vee a) \vee a) \quad \neg(\neg(\neg(\neg a \vee b) \vee a) \vee a)}{(\neg(\neg(\neg(\neg 0 \vee b) \vee 0) \vee 0)) \vee (\neg(\neg(\neg(\neg 1 \vee b) \vee 1) \vee 1))} \text{CA (on a)}}{0 \vee (\neg(\neg(\neg b \vee 1) \vee 1))} \text{SIMP}}{\neg(\neg(\neg b \vee 1) \vee 1)} \text{SIMP}$$

Note that, if we are being pedantic, we cannot simplify further as the simplification rules only cover cases where a literal is on the left.

$$\frac{\frac{\neg(\neg(\neg b \vee 1) \vee 1) \quad \neg(\neg(\neg b \vee 1) \vee 1)}{(\neg(\neg(\neg 0 \vee 1) \vee 1)) \vee (\neg(\neg(\neg 1 \vee 1) \vee 1))} \text{CA (on b)}}{0 \vee 0} \text{SIMP}}{0} \text{SIMP}$$

2.

$$\frac{\frac{\neg(\neg\neg(a \wedge b) \vee (\neg a \vee \neg b)) \quad \neg(\neg\neg(a \wedge b) \vee (\neg a \vee \neg b))}{(\neg(\neg\neg(0 \wedge b) \vee (\neg 0 \vee \neg b))) \vee (\neg(\neg\neg(1 \wedge b) \vee (\neg 1 \vee \neg b)))} \text{CA (on a)}}{0 \vee (\neg(\neg\neg b \vee \neg b))} \text{SIMP}}{\neg(\neg\neg b \vee \neg b)} \text{SIMP}$$

$$\frac{\frac{\neg(\neg\neg b \vee \neg b) \quad \neg(\neg\neg b \vee \neg b)}{(\neg(\neg\neg 0 \vee \neg 0)) \vee (\neg(\neg\neg 1 \vee \neg 1))} \text{CA (on b)}}{0} \text{SIMP}$$

3.

$$\frac{\frac{\neg(\neg((\neg\neg a \vee b) \wedge (\neg a \vee b)) \vee b) \quad \neg(\neg((\neg\neg a \vee b) \wedge (\neg a \vee b)) \vee b)}{(\neg(\neg((\neg\neg 0 \vee b) \wedge (\neg 0 \vee b)) \vee b)) \vee (\neg(\neg((\neg\neg 1 \vee b) \wedge (\neg 1 \vee b)) \vee b))} \text{CA (on a)}}{\neg(\neg(b \wedge 1) \vee b) \vee \neg(\neg b \vee b)} \text{SIMP}$$

$$\frac{\frac{\neg(\neg(b \wedge 1) \vee b) \vee \neg(\neg b \vee b) \quad \neg(\neg(b \wedge 1) \vee b) \vee \neg(\neg b \vee b)}{(\neg(\neg(0 \wedge 1) \vee 0) \vee \neg(\neg 0 \vee 0)) \vee (\neg(\neg(1 \wedge 1) \vee 1) \vee \neg(\neg 1 \vee 1))} \text{CA (on b)}}{0} \text{SIMP}$$

Exercise 3 (if-then-else).

1. Let's first express f1 and f2 as propositional formulas. **if x then y else z** means that when x is true then y is true and when x is not true z is true. In propositional logic this becomes $(x \rightarrow y) \wedge (\neg x \rightarrow z)$. By repeating the process and reducing them to CNF, we get the following formulas for f1 and f2:

$$\begin{aligned} \text{f1}(a, b, c) &\equiv ((a \vee b) \rightarrow (((b \wedge \neg a) \rightarrow (b \wedge c)) \wedge (\neg(b \wedge \neg a) \rightarrow (\neg b \wedge c)))) \wedge (\neg(a \vee b) \rightarrow c) \\ &\equiv (\neg(a \vee b) \vee ((\neg(b \wedge \neg a) \vee (b \wedge c)) \wedge ((b \wedge \neg a) \vee (\neg b \wedge c)))) \wedge ((a \vee b) \vee c) \\ &\equiv (\neg a \vee \neg b) \wedge c \end{aligned}$$

$$\begin{aligned} \text{f2}(a, b, c) &\equiv (c \rightarrow (a \rightarrow \neg b) \wedge (\neg a \rightarrow 1)) \wedge (\neg c \rightarrow 0) \\ &\equiv (\neg c \vee (\neg a \vee \neg b) \wedge (a \vee 1)) \wedge (c \vee 0) \\ &\equiv (\neg a \vee \neg b) \wedge c \end{aligned}$$

which proves that they do always produce the same output. One could also have computed the truth tables of each formula and check that they were equal.

2. Since $a \uparrow b \equiv \text{if (if a then b else false) then false else true}$, and any formula can be written with Sheffer strokes, any formula can also be written only using **if then else**.

By noting that $\varphi \equiv \text{if } a \text{ then } \varphi[a := 1] \text{ else } \varphi[a := 0]$ where $a \in \text{FV}(\varphi)$ and by applying the identity recursively we can express any formula with **if then else** with only one variable in the condition. The proof goes by induction on the number of variables in the formula, and by noting that if φ has n variables and $a \in \text{FV}(\varphi)$, then $\varphi[a := 0]$ and $\varphi[a := 1]$ contain only $n - 1$ variables.

Exercise 4 (Propositional Tautologies). Remember that for any formulas P , Q , and R , the formula $P \rightarrow Q \rightarrow R$ is parsed as $P \rightarrow (Q \rightarrow R)$ and is equivalent to $(P \wedge Q) \rightarrow R$.

1. If $P \wedge Q$ is true, then so is Q .
2. We consider two cases. If $P \rightarrow Q$ holds, then we are done, otherwise we have $P \wedge \neg Q$, and we are done as well.
3. Counterexample: $P = \top$, $Q = \perp$, $R = \perp$.
4. Counterexample: $P = \top$, $Q = \top$, $R = \perp$.
5. If R and $\neg R$ are true, then we have a contradiction and Q is true as well.
6. A formula always implies itself: $(P \rightarrow Q) \rightarrow (P \rightarrow Q)$.
7. (Peirce's law) We consider two cases. If P is true, then the whole formula is true. Otherwise, $(P \rightarrow Q)$ is true, meaning the implication $((P \rightarrow Q) \rightarrow P)$ is false. This makes the whole formula true as well.
8. Counterexample: $P = \top$, $Q = \perp$.
9. Assume $(\neg Q \rightarrow \neg P)$ and P are true. Assume now (by contradiction) that Q is false. Then, we would have that P is false, which is a contradiction. Therefore Q is true and so is the whole formula.
10. Counterexample: $P = \top$, $Q = \top$, $R = \perp$.
11. Assume the three disjunctions are true. We want to show that P is true as well. Consider the first two disjunctions (the third one is not needed). In either of them, if the left-hand-side (P) is true, then we are done. Otherwise, it means their right-hand-sides Q and $\neg Q$ are both true, which is a contradiction.
12. Counterexample: $P = \top$, $Q = \perp$.
13. Assume $\neg(P \wedge Q)$ and P are both true. If Q is true as well, then, we have $P \wedge Q$ is true, which is a contradiction. Therefore Q is false, which is what we needed to prove.

1 Transition Systems and Invariants

Exercise 5 (Special Invariants).

1. S is an inductive invariant. It is the largest among all invariants.

Solution: We have $S \subseteq S$, $I \subseteq S$, and for every $s, s' \in S$ and $a \in A$ such that $(s, a, s') \in r$, $s' \in S$. Therefore, S is an inductive invariant.

Moreover, S is the largest among all invariants because any invariant needs to be a subset of S .

2. $Reach(M)$ is an inductive invariant.

Solution: First, we have $Reach(M) \subseteq S$ and $I \subseteq Reach(M)$.

Then, we want to show that for all $s, s' \in S$ and $a \in A$ such that $(s, a, s') \in r$, if $s \in Reach(M)$, then $s' \in Reach(M)$.

Let $s, s' \in S$ and $a \in A$ such that $(s, a, s') \in r$. Assume $s \in Reach(M)$. By definition of $Reach$, there exists a trace $(s_0, a_0, s_1, \dots, a_n, s_n) \in Traces(M)$ such that $s_n = s$. By definition of $Traces$, we also have $(s_0, a_0, s_1, \dots, a_n, s, a, s') \in Traces(M)$, which proves that $s' \in Reach(M)$.

3. $Reach(M)$ is the smallest of all possible invariants.

Solution: By definition, any invariant is a superset of $Reach(M)$.

Exercise 6 (Closure of Invariants).

1. union

Solution: Let P_1 and P_2 be two invariants, i.e. $Reach(M) \subseteq P_1$, and $Reach(M) \subseteq P_2$. We have $Reach(M) \subseteq P_1 \cup P_2$, therefore $P_1 \cup P_2$ is also an invariant.

2. intersection

Solution: Let P_1 and P_2 be two invariants, i.e. $Reach(M) \subseteq P_1$, and $Reach(M) \subseteq P_2$. We have $Reach(M) \subseteq P_1 \cap P_2$, because every state in $Reach(M)$ belongs to both P_1 and P_2 by assumption. Therefore $P_1 \cap P_2$ is also an invariant.

3. complement with respect to S

Solution: Let P be an invariant, i.e. $Reach(M) \subseteq P$. We don't necessarily have $S \setminus Reach(M) \subseteq P$. For instance, we can make a machine where $Reach(M) = \emptyset$ (no initial states), with S a non-empty set of states, and define $P = \emptyset$.

4. operation $f : 2^S \rightarrow 2^S$ defined by $f(X) = Reach(M) \cup (S \setminus X)$

Solution: For any set X (not just invariants), we have $Reach(M) \subseteq f(X)$, therefore $f(X)$ is an invariant.

Answer the same questions about all *inductive invariants* of M .

1. union

Solution: Let P_1 and P_2 be two inductive invariants. We show that $P_1 \cup P_2$ is also an inductive invariant.

First, we have $I \subseteq P_1 \cup P_2$ (because e.g. $I \subseteq P_1$).

Second, let $s, s' \in S$ and $a \in A$ such that $(s, a, s') \in r$. Assume $s \in P_1 \cup P_2$. We consider two cases. If $s \in P_1$, we use the fact that P_1 is an inductive invariant to conclude that $s' \in P_1$, and therefore $s' \in P_1 \cup P_2$. The second case is similar.

2. intersection

Solution: Let P_1 and P_2 be two inductive invariants. We show that $P_1 \cap P_2$ is also an inductive invariant.

First, we have $I \subseteq P_1 \cap P_2$ (because elements of I are both in P_1 and in P_2).

Second, let $s, s' \in S$ and $a \in A$ such that $(s, a, s') \in r$. Assume $s \in P_1 \cap P_2$, i.e. $s \in P_1$, and $s \in P_2$. Since P_1 is an inductive invariant, we deduce that $s' \in P_1$, and similarly since P_2 is an inductive invariant, we deduce that $s' \in P_2$. Therefore $s' \in P_1 \cap P_2$.

3. complement with respect to S

Solution: Same counter-example as above.

4. operation $f : 2^S \rightarrow 2^S$ defined by $f(X) = \text{Reach}(M) \cup (S \setminus X)$

Solution: When X is an inductive invariant, $f(X)$ is not necessarily an inductive invariant. Consider a machine with two states $S = \{s_1, s_2\}$ and no initial state. Let there be a transition from s_1 to s_2 (with some letter), and no other transitions. The set $X = \{s_2\}$ is an inductive invariant. However, $f(X) = \{s_1\}$ is not (because of the transition from s_1 to s_2).

2 Relations

Exercise 7 (Relation Identities). In the solutions, some equivalences are easier to read from bottom to top (and some from top to bottom).

1. $(X \bullet r_1) \bullet r_2 = X \bullet (r_1 \circ r_2)$

Solution: Let $z \in A$. We have:

$$\begin{aligned}
z \in (X \bullet r_1) \bullet r_2 &\Leftrightarrow \exists y \in X \bullet r_1. (y, z) \in r_2 \\
&\Leftrightarrow \exists y \in A. y \in X \bullet r_1 \wedge (y, z) \in r_2 \\
&\Leftrightarrow \exists y \in A. \exists x \in X. (x, y) \in r_1 \wedge (y, z) \in r_2 \\
&\Leftrightarrow \exists x \in X. \exists y \in A. (x, y) \in r_1 \wedge (y, z) \in r_2 \\
&\Leftrightarrow \exists x \in X. (x, z) \in r_1 \circ r_2 \\
&\Leftrightarrow z \in X \bullet (r_1 \circ r_2)
\end{aligned}$$

2. $(r \cup s) \circ t = (r \circ t) \cup (s \circ t)$

Solution: Let $x, z \in A$. We have:

$$\begin{aligned}
(x, z) \in (r \cup s) \circ t &\Leftrightarrow \exists y \in A. (x, y) \in r \cup s \wedge (y, z) \in t \\
&\Leftrightarrow \exists y \in A. ((x, y) \in r \vee (x, y) \in s) \wedge (y, z) \in t \\
&\Leftrightarrow \exists y \in A. ((x, y) \in r \wedge (y, z) \in t) \vee ((x, y) \in s \wedge (y, z) \in t) \\
&\Leftrightarrow (\exists y \in A. (x, y) \in r \wedge (y, z) \in t) \vee (\exists y \in A. (x, y) \in s \wedge (y, z) \in t) \\
&\Leftrightarrow (x, z) \in r \circ t \vee (x, z) \in s \circ t \\
&\Leftrightarrow (x, z) \in (r \circ t) \cup (s \circ t)
\end{aligned}$$

3. $(r \cap s) \circ t = (r \circ t) \cap (s \circ t)$

Solution: A counterexample:

$$r = \{(0, 1)\}, s = \{(0, 2)\} \text{ and } t = \{(1, 0), (2, 0)\}.$$

We have $(r \cap s) \circ t = \emptyset$, while $(r \circ t) \cap (s \circ t) = \{(0, 0)\}$.

4. $(r_1 \circ r_2)^{-1} = (r_2^{-1} \circ r_1^{-1})$

Solution: Let $x, z \in A$. We have:

$$\begin{aligned}
(z, x) \in (r_1 \circ r_2)^{-1} &\Leftrightarrow (x, z) \in r_1 \circ r_2 \\
&\Leftrightarrow \exists y \in A. (x, y) \in r_1 \wedge (y, z) \in r_2 \\
&\Leftrightarrow \exists y \in A. (y, x) \in r_1^{-1} \wedge (z, y) \in r_2^{-1} \\
&\Leftrightarrow \exists y \in A. (z, y) \in r_2^{-1} \wedge (y, x) \in r_1^{-1} \\
&\Leftrightarrow (z, x) \in r_2^{-1} \circ r_1^{-1}
\end{aligned}$$

5. $X \bullet r = \text{ran}(\Delta_X \circ r)$

Solution: Let $y \in A$. We have:

$$\begin{aligned}
y \in X \bullet r &\Leftrightarrow \exists x \in X. (x, y) \in r \\
&\Leftrightarrow \exists x \in A. x \in X \wedge (x', y) \in r \\
&\Leftrightarrow \exists x \in A. \exists x' \in A. x = x' \wedge x \in X \wedge (x', y) \in r \\
&\Leftrightarrow \exists x \in A. \exists x' \in A. (x, x') \in \Delta_X \wedge (x', y) \in r \\
&\Leftrightarrow \exists x \in A. (x, y) \in \Delta_X \circ r \\
&\Leftrightarrow y \in \text{ran}(\Delta_X \circ r)
\end{aligned}$$

6. If $r_1 \subseteq r'_1$ then $r_1 \circ r_2 \subseteq r'_1 \circ r_2$ and $r_2 \circ r_1 \subseteq r_2 \circ r'_1$.

Solution: Let $x, z \in A$. We have:

$$\begin{aligned}
(x, z) \in r_1 \circ r_2 &\Leftrightarrow \exists y \in A. (x, y) \in r_1 \wedge (y, z) \in r_2 \\
&\Rightarrow \exists y \in A. (x, y) \in r'_1 \wedge (y, z) \in r_2 \text{ because } r_1 \subseteq r'_1 \\
&\Leftrightarrow (x, z) \in r'_1 \circ r_2
\end{aligned}$$

Also,

$$\begin{aligned}
(x, z) \in r_2 \circ r_1 &\Leftrightarrow \exists y \in A. (x, y) \in r_2 \wedge (y, z) \in r_1 \\
&\Rightarrow \exists y \in A. (x, y) \in r_2 \wedge (y, z) \in r'_1 \text{ because } r_1 \subseteq r'_1 \\
&\Leftrightarrow (x, z) \in r_2 \circ r'_1
\end{aligned}$$

7. If $r_1 \subseteq r'_1$ then $r_1 \cup r_2 \subseteq r'_1 \cup r_2$ and $r_2 \cup r_1 \subseteq r_2 \cup r'_1$.

Solution: Let $x, y \in A$. We have:

$$\begin{aligned}
(x, y) \in r_1 \cup r_2 &\Leftrightarrow (x, y) \in r_1 \vee (x, y) \in r_2 \\
&\Rightarrow (x, y) \in r'_1 \vee (x, y) \in r_2 \text{ because } r_1 \subseteq r'_1 \\
&\Leftrightarrow (x, y) \in r'_1 \cup r_2
\end{aligned}$$

Also,

$$\begin{aligned}
(x, y) \in r_2 \cup r_1 &\Leftrightarrow (x, y) \in r_2 \vee (x, y) \in r_1 \\
&\Rightarrow (x, y) \in r_2 \vee (x, y) \in r'_1 \text{ because } r_1 \subseteq r'_1 \\
&\Leftrightarrow (x, y) \in r_2 \cup r'_1
\end{aligned}$$

Exercise 8 (Transitive relations).

$$\begin{aligned}
r \circ r \subseteq r &\Leftrightarrow \forall x, z \in A. (x, z) \in r \circ r \Rightarrow (x, z) \in r \\
&\Leftrightarrow \forall x, z \in A. (\exists y \in A. (x, y) \in r \wedge (y, z) \in r) \Rightarrow (x, z) \in r \\
&\Leftrightarrow \forall x, z \in A. \forall y \in A. ((x, y) \in r \wedge (y, z) \in r) \Rightarrow (x, z) \in r \\
&\Leftrightarrow \forall x, y, z \in A. ((x, y) \in r \wedge (y, z) \in r) \Rightarrow (x, z) \in r \\
&\Leftrightarrow r \text{ is transitive}
\end{aligned}$$

Exercise 9 (Symmetric relations). First, since the composition of two symmetric relations is symmetric, we can show by induction on $n \geq 0$ that for any symmetric relation r , r^n is symmetric.

Second, taking the union of symmetric relations gives a symmetric relation, which means that for any symmetric relation r , r^* is also symmetric (using the previous point).

We can establish that $r \cup r^{-1}$ is a symmetric relation, and using the second point, that $(r \cup r^{-1})^*$ is also symmetric.

Finally, let $x, t \in A$. We have:

$$\begin{aligned}
(x, t) \in r^{-1} \circ (r \cup r^{-1})^* \circ r &\Leftrightarrow \exists y \in A. \exists z \in A. (x, y) \in r^{-1} \wedge (y, z) \in (r \cup r^{-1})^* \wedge (z, t) \in r \\
&\Leftrightarrow \exists z \in A. \exists y \in A. (t, z) \in r^{-1} \wedge (z, y) \in (r \cup r^{-1})^* \wedge (y, x) \in r \\
&\Leftrightarrow (t, x) \in r^{-1} \circ (r \cup r^{-1})^* \circ r
\end{aligned}$$

To go from the first to the second line, we use the fact that $(r \cup r^{-1})^*$ is symmetric.

Exercise 10 (Transitive closure).

(i) $(r \cup r^{-1})^*$ is an equivalence relation,

Solution: We have three things to prove:

1. (Reflexivity) Let $x \in A$. We have $(x, x) \in (r \cup r^{-1})^*$ because $\Delta_A \subseteq (r \cup r^{-1})^*$.
2. (Symmetry) As seen in the previous section, $(r \cup r^{-1})^*$ is symmetric.
3. (Transitive) Let $(x, y) \in (r \cup r^{-1})^*$ and $(y, z) \in (r \cup r^{-1})^*$. By definition, there exist $n, m \geq 0$ such that $(x, y) \in (r \cup r^{-1})^n$ and $(y, z) \in (r \cup r^{-1})^m$. Therefore $(x, z) \in (r \cup r^{-1})^n \circ (r \cup r^{-1})^m = (r \cup r^{-1})^{n+m}$. This shows that $(x, z) \in (r \cup r^{-1})^*$.

(ii) if s is an equivalence relation containing r , then $(r \cup r^{-1})^* \subseteq s$.

Solution: Let s be an equivalence relation containing r . We prove by induction over $n \in \mathbb{N}$ that $(r \cup r^{-1})^n \subseteq s$.

1. (Case $n = 0$) $(r \cup r^{-1})^0 = \Delta_A \subseteq s$ because s is reflexive.
2. (Case $n = n' + 1$) Assume by induction that $(r \cup r^{-1})^{n'} \subseteq s$. We know that $r \subseteq s$, and that s is symmetric, therefore $r^{-1} \subseteq s$. Thus, we also have: $r \cup r^{-1} \subseteq s$.
Finally, we have $(r \cup r^{-1})^n = (r \cup r^{-1})^{n'} \circ (r \cup r^{-1}) \subseteq s \circ s \subseteq s$ (because s is transitive), which concludes the proof.

3 Finite State Machines with Boolean Variables

Exercise 11 (Finite State Machines with Boolean Variables). No. We can establish the following inductive invariant of the configurations of the FSM:

$$\begin{aligned}
 P = \{ & ((x, y), m) \mid (x \geq 1 \Rightarrow m(B)) \wedge \\
 & (y \geq 1 \Rightarrow m(A)) \wedge \\
 & ((x, y) = (3, 2) \Rightarrow m(C)) \wedge \\
 & ((x, y) = (2, 3) \Rightarrow \neg m(C)) \wedge \\
 & (x, y) \neq (3, 3) \}
 \end{aligned}$$

Part B - More Logic

Exercise 12 (Traces).

1. Since there are n variables, which can take two values, there are 2^n possible states. We can construct a trace passing through each state exactly once.
2. As in class, $\bar{s} = (s_1, \dots, s_n)$ and exponents, e.g. $\bar{s}^k$, are used to represent variables in different states (i.e. $\bar{s}^k = (s_1^k, \dots, s_n^k)$).

(a) Noting

$$\bar{v}^k = (s_1^k, \dots, s_n^k, a_1^k, \dots, a_m^k, x_1^k, \dots, x_\ell^k),$$

$$F_{(=k)} = \exists \bar{v}^0, \dots, \bar{v}^{k-1}, \bar{s}^k. \text{Init} [\bar{s} := \bar{s}^0] \wedge \left(\bigwedge_{i=0}^{k-1} R_i \right) \wedge (\bar{s}^k = \bar{s}).$$

Another (equivalent) solution:

$$\exists \bar{v}^0, \dots, \bar{v}^{k-1}.$$

$$\text{Init} [\bar{s} := \bar{s}^0] \wedge \left(\bigwedge_{i=0}^{k-2} R_i \right) \wedge R \left[(\bar{s}, \bar{a}, \bar{x}, \bar{s}') := (\bar{s}^{k-1}, \bar{a}^{k-1}, \bar{x}^{k-1}, \bar{s}) \right].$$

(b) $F_{(\leq k)} = \bigvee_{i=0}^k F_{(=i)}.$

(c) The “intuitive” solution is $F_{(\leq \infty)}$. However, this formula is infinite. Using question (1), it can be reduced to $F_{(\text{reachable})} = F_{(\leq 2^n - 1)}$.

(d) $F_{(=k, \text{Err})} = \exists \bar{v}^0, \dots, \bar{v}^{k-1}, \bar{s}^k. (\bar{s}^0 = \bar{s}) \wedge \left(\bigwedge_{i=0}^{k-1} R_i \right) \wedge E [\bar{s} := \bar{s}^k]$

(e) $F_{(\leq k, \text{Err})} = \bigvee_{i=0}^k F_{(=i, \text{Err})}$

(f) $F_{(\text{stuck})} = \neg \exists \bar{a}, \bar{x}, \bar{s}'. R$

3. The formulas are:

(a) $\neg \exists \bar{s}. F_{(\text{reachable})} \wedge E$

(b) $\neg \exists \bar{s}. F_{(\text{reachable})} \wedge F_{(\text{stuck})}$

(c) $\forall \bar{s}. F_{(\leq k)}$

(d) Note that we are only allowed to quantify on propositional variables. As such, a formula such as

$$\forall k. \exists \bar{s}. F_{(=k)} \wedge \neg F_{(\text{stuck})}$$

is not allowed, as k ranges (implicitly) over $\mathbb{N}$.

Consider instead

$$\exists \bar{s}. F_{(=2^n)}$$

if such a state ($\bar{s}$) exists, the underlying trace contains $2^n + 1$ states (since it is 2^n steps long). By the pigeon-hole principle, one of these states must appear twice in the trace. This means that there is a part of the trace which can be repeated infinitely (“pumped”), implying the existence of an infinite trace (see the proof of the [pumping lemma](#) for regular languages which relies on the same principle).

Exercise 13 (Tseytin Transformation).

1. Any expression of the type $y_k \leftrightarrow x_i * x_j$ is easily convertible to a CNF.

$$\begin{aligned}
y_k \leftrightarrow x_i \vee x_j &\equiv (y_k \rightarrow x_i \vee x_j) \wedge (x_i \vee x_j \rightarrow y_k) \\
&\equiv (\neg y_k \vee x_i \vee x_j) \wedge (\neg x_i \vee y_k) \wedge (\neg x_j \vee y_k) \\
y_k \leftrightarrow x_i \wedge x_j &\equiv (y_k \rightarrow x_i \wedge x_j) \wedge (x_i \wedge x_j \rightarrow y_k) \\
&\equiv (\neg y_k \vee x_i) \wedge (\neg y_k \vee x_j) \wedge (\neg x_i \vee \neg x_j \vee y_k) \\
y_k \leftrightarrow x_i \rightarrow x_j &\equiv y_k \leftrightarrow \neg x_i \vee x_j \\
&\equiv (\neg y_k \vee \neg x_i \vee x_j) \wedge (x_i \vee y_k) \wedge (\neg x_j \vee y_k) \\
y_k \leftrightarrow \neg x_i &\equiv (y_k \rightarrow \neg x_i) \wedge (\neg x_i \rightarrow y_k) \\
&\equiv (\neg y_k \vee \neg x_i) \wedge (x_i \vee y_k)
\end{aligned}$$

Let’s first define the mapping between sub formulas and fresh variables starting from the leaves:

$$\begin{aligned}
y_1 &\leftrightarrow \neg b \\
y_2 &\leftrightarrow a \vee y_1 \\
y_3 &\leftrightarrow b \rightarrow c \\
y_4 &\leftrightarrow \neg y_3 \\
y_5 &\leftrightarrow y_2 \wedge y_4 \\
y_6 &\leftrightarrow y_5 \rightarrow a \\
y_7 &\leftrightarrow \neg y_6
\end{aligned}$$

The Tseytin transformation of the formula is therefore:

$$\begin{aligned}
&(y_1 \leftrightarrow \neg b) \wedge (y_2 \leftrightarrow a \vee y_1) \wedge (y_3 \leftrightarrow b \rightarrow c) \wedge (y_4 \leftrightarrow \neg y_3) \wedge \\
&(y_5 \leftrightarrow y_2 \wedge y_4) \wedge (y_6 \leftrightarrow y_5 \rightarrow a) \wedge (y_7 \leftrightarrow \neg y_6) \wedge y_7 \\
= &(\neg y_1 \vee \neg b) \wedge (b \vee y_1) \wedge (\neg y_2 \vee a \vee y_1) \wedge (\neg a \vee y_2) \wedge (\neg y_1 \vee y_2) \wedge \\
&(\neg y_3 \vee \neg b \vee c) \wedge (b \vee y_3) \wedge (\neg c \vee y_3) \wedge (\neg y_4 \vee \neg y_3) \wedge (y_3 \vee y_4) \wedge \\
&(\neg y_5 \vee y_2) \wedge (\neg y_5 \vee y_4) \wedge (\neg y_2 \vee \neg y_4 \vee y_5) \wedge (\neg y_6 \vee \neg y_5 \vee a) \wedge \\
&(y_5 \vee y_6) \wedge (\neg a \vee y_6) \wedge (\neg y_6 \vee \neg y_7) \wedge (y_6 \vee y_7) \wedge y_7
\end{aligned}$$

2. We first show that they might not be equivalent by providing a small counterexample. As written above, the Tseytin transform of the formula $\neg x$ is $(y \leftrightarrow \neg x) \wedge y$. The value of $\neg x$ when $x = 0$ is 1, whereas the value of its Tseytin transformation when $x = 0, y = 0$, is 0. Therefore the two formulas are not equivalent.

Let's now prove that a formula is satisfiable if and only if its Tseytin transformation is satisfiable as well.

Let φ be a formula and $V = \text{FV}(\varphi)$ its set of free variables. One can notice (and prove by induction) that the Tseytin transformation of φ is equivalent to:

$$y_n \wedge \bigwedge_{i=1}^n (y_i \leftrightarrow \psi_i)$$

where $\{\psi_i\}_{0 \leq i \leq n}$ are all the (non-atomic) subformulas of φ ; in particular, $\psi_n = \varphi$.

φ is satisfiable iff there exists an assignment α such that $\varphi[V := \alpha] = 1$. However, note that the formula $(y_i \leftrightarrow \psi_i)$ is true if and only if one assigns the truth value of $\psi_i[V := \alpha]$ to y_i . We can construct a new assignment α' as follows:

$$\alpha'(v) = \begin{cases} \psi_i[V := \alpha] & \text{if } v = y_i \text{ for some } 0 \leq i \leq n \\ \alpha(v) & \text{otherwise} \end{cases}$$

Convince yourself that this is a satisfying assignment for both the original formula φ and its Tseytin transformation. Conversely, let α' be a satisfying assignment for the Tseytin transformation of φ . This is already a satisfying assignment for φ .

Thus, there exists a satisfying assignment for φ if and only if there exists a satisfying assignment for its Tseytin transformation. Therefore, they are equisatisfiable.

3. F is a valid formula if and only if $\neg F$ is false for any variable assignment, i.e. it is unsatisfiable. A formula is unsatisfiable if and only if its Tseytin transformation is unsatisfiable as well (by the previous question). Therefore F is a valid formula if and only if the Tseytin transformation of $\neg F$ is unsatisfiable.

By induction on the structure of the formula, one can show that formula size increases by at most a constant amount for each "building block" of the formula (literal or binary operator). Therefore the size of the Tseytin transformation is in $O(n)$.

Exercise 14 (SAT solving). By taking the CNF found in the above exercise and applying resolution, we obtain the following derivation trees:

$$\begin{array}{c}
\frac{\frac{\neg a \vee y_6 \quad \frac{y_7 \quad \neg y_6 \vee \neg y_7}{\neg y_6}}{\neg a \text{ (1)}} \quad \frac{\frac{b \vee y_3 \quad \frac{\neg y_3 \vee \neg y_4}{\neg y_3}}{b \text{ (2)}} \quad \frac{\frac{y_4 \vee \neg y_5 \quad y_5}{y_4} \quad \frac{y_7 \quad \neg y_6 \vee \neg y_7}{y_5 \vee y_6 \quad \neg y_6}}{\neg y_1 \vee \neg b \quad \neg y_2 \vee a \vee y_1 \quad \neg y_5 \vee y_2 \quad y_5 \quad y_7 \quad \neg y_6 \vee \neg y_7}{\neg b \vee a \vee \neg y_2 \quad y_2 \quad y_5 \quad y_6 \quad \neg y_6}}{\neg b \vee a \text{ (3)}} \quad \frac{\frac{b \text{ (2)} \quad \neg b \vee a \text{ (3)}}{\neg b} \quad \neg a \text{ (1)}}{F}
\end{array}$$

Exercise 15 (First-order formulas). The proofs are not given for all cases. The omitted cases are similar to those already given.

1. $A : \forall x.(P(x) \vee Q(x))$
 $B : (\forall x.P(x)) \vee (\forall x.Q(x))$

(a) $A \models B$: ✗. Consider $(\mathbb{N}, e)$ with

$$\begin{aligned}
e(P) &= \{n \in \mathbb{N} \mid n \bmod 2 = 0\} \\
e(Q) &= \{n \in \mathbb{N} \mid n \bmod 2 = 1\}
\end{aligned}$$

(b) $B \models A$: ✓. Consider any structure (D, e) such that $(D, e) \models B$. We have $\llbracket B \rrbracket_{(D, e)} = 1$. By definition of $\llbracket \cdot \vee \cdot \rrbracket_{(D, e)}$, we know that either $\llbracket \forall x.P(x) \rrbracket_{(D, e)} = 1$ or $\llbracket \forall x.Q(x) \rrbracket_{(D, e)} = 1$. Pick any case, the other is identical. Supposing $\llbracket \forall x.P(x) \rrbracket_{(D, e)} = 1$, we know that for every $d \in D$, $\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 1$. However, this implies that for every $d \in D$, $\llbracket P(x) \vee Q(x) \rrbracket_{(D, e[x:=d])} = 1$, and thus $\llbracket \forall x.(P(x) \vee Q(x)) \rrbracket_{(D, e)} = 1$. Therefore, $(D, e) \models A$ as well.

2. $A : \exists x.(P(x) \vee Q(x))$
 $B : (\exists x.P(x)) \vee (\exists x.Q(x))$

(a) $A \models B$: ✓. Consider any structure (D, e) such that $(D, e) \models A$. We have $\llbracket A \rrbracket_{(D, e)} = 1$. By definition of $\llbracket \cdot \rrbracket_{(D, e)}$, there exists $d \in D$ such that $\llbracket P(x) \vee Q(x) \rrbracket_{(D, e[x:=d])} = 1$. Thus, either $\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 1$ or $\llbracket Q(x) \rrbracket_{(D, e[x:=d])} = 1$. In the first case, $\llbracket \exists x.P(x) \rrbracket_{(D, e)} = 1$, and in the second case, $\llbracket \exists x.Q(x) \rrbracket_{(D, e)} = 1$. In either case, $(D, e) \models B$.

(b) $B \models A$: ✓. Proof omitted.

3. $A : \forall x.(P(x) \wedge Q(x))$
 $B : (\forall x.P(x)) \wedge (\forall x.Q(x))$

(a) $A \models B$: ✓. Proof omitted.

(b) $B \models A$: ✓. Proof omitted.

4. $A : \exists x.(P(x) \wedge Q(x))$
 $B : (\exists x.P(x)) \wedge (\exists x.Q(x))$

(a) $A \models B$: ✓. Proof omitted.

(b) $B \models A$: ✗. Consider $(\mathbb{N}, e)$ with

$$e(P) = \{0\}$$

$$e(Q) = \{1\}$$

5. $A : \forall x.(P(x) \rightarrow Q(x))$
 $B : (\forall x.P(x)) \rightarrow (\forall x.Q(x))$

(a) $A \models B$: ✓. Consider any structure (D, e) such that $(D, e) \models A$. We have $\llbracket A \rrbracket_{(D, e)} = 1$, i.e., for every $d \in D$, $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D, e[x:=d])} = 1$. Note that $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D, e[x:=d])} = \llbracket \neg P(x) \vee Q(x) \rrbracket_{(D, e)}$, and thus for each d :

$$\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 0 \text{ or } \llbracket Q(x) \rrbracket_{(D, e[x:=d])} = 1 \quad (1)$$

Assume, towards a contradiction, that $\llbracket B \rrbracket_{(D, e)} = 0$. We must have

$$\llbracket \forall x.P(x) \rrbracket_{(D, e)} = 1 \text{ and } \llbracket \forall x.Q(x) \rrbracket_{(D, e)} = 0 \quad (2)$$

The left conjunct of Equation 2 tells us that for every $d \in D$, $\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 1$. By Equation 1, this implies that for every $d \in D$, $\llbracket Q(x) \rrbracket_{(D, e[x:=d])} = 1$. This contradicts the right conjunct of Equation 2. Therefore, $\llbracket B \rrbracket_{(D, e)} = 1$, and thus $(D, e) \models B$.

(b) $B \models A$: ✗. Consider $(\mathbb{N}, e)$ with

$$e(P) = \{0\}$$

$$e(Q) = \{1\}$$

6. $A : \exists x.(P(x) \rightarrow Q(x))$
 $B : (\exists x.P(x)) \rightarrow (\exists x.Q(x))$

(a) $A \models B$: ✗. Consider $(\mathbb{N}, e)$ with

$$e(P) = \{0\}$$

$$e(Q) = \emptyset$$

Take $x = 1$ as a witness for A .

- (b) $B \models A$: ✓. Consider any structure (D, e) such that $(D, e) \models B$. We have $\llbracket B \rrbracket_{(D, e)} = 1$. As before, by expanding the implication, we must have either $\llbracket \exists x.P(x) \rrbracket_{(D, e)} = 0$ or $\llbracket \exists x.Q(x) \rrbracket_{(D, e)} = 1$. In the first case, we have that for every $d \in D$, $\llbracket P(x) \rrbracket_{(D, e[x:=d])} = 0$. This implies that for every $d \in D$, $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D, e[x:=d])} = 1$, and thus in particular $\llbracket \exists x.P(x) \rightarrow Q(x) \rrbracket_{(D, e)} = 1$. In the second case, there exists $d \in D$ such that $\llbracket Q(x) \rrbracket_{(D, e[x:=d])} = 1$. Applying the interpretation of implication again, we have that $\llbracket P(x) \rightarrow Q(x) \rrbracket_{(D, e[x:=d])} = 1$, and thus $\llbracket \exists x.P(x) \rightarrow Q(x) \rrbracket_{(D, e)} = 1$. Thus, in either case, $(D, e) \models A$.

Exercise 16 (First-order structures).

1. Let $I = (D, \alpha)$ be an interpretation of $\mathcal{L}$, for which $\alpha(B) = b$ and $\alpha(T) = t$. For notational convenience we will denote $\alpha(L)$ by L_α . The consequences of axioms in S are:

$$\begin{aligned} \llbracket \text{Min} \rrbracket_I = 1 &\implies L_\alpha(b, b), L_\alpha(b, u) \text{ and } L_\alpha(b, t) \\ \llbracket \text{Max} \rrbracket_I = 1 &\implies L_\alpha(b, t), L_\alpha(u, t) \text{ and } L_\alpha(t, t) \\ \llbracket \text{Tot} \rrbracket_I = 1 &\implies L_\alpha(u, u) \text{ (by choosing } x = u \text{ and } y = u) \\ \llbracket \text{Tra} \rrbracket_I = 1 &\implies (L_\alpha(t, b) \implies L_\alpha(t, u)) \end{aligned}$$

This greatly reduces the number of choices. We can fill out the partial table for L_α :

L_α	b	u	t
b	1	1	1
u	?	1	1
t	?	?	1

Finally, based on **Tra**, L_α is restricted to be either:

a)	L_α	b	u	t
	b	1	1	1
	u	1	1	1
c)	L_α	b	u	t
	b	1	1	1
	u	0	1	1
b)	L_α	b	u	t
	b	1	1	1
	u	1	1	1
d)	L_α	b	u	t
	b	1	1	1
	u	0	1	1

All the options satisfy reflexivity and transitivity as a consequence of the axioms. However, only d) satisfies the anti symmetry property. Therefore only d) is a partial order.

2.

a)	$E_{\alpha'}$	b	u	t
	b	1	1	1
	u	1	1	1
	t	1	1	1

b)	$E_{\alpha'}$	b	u	t
	b	1	1	0
	u	1	1	0
	t	0	0	1

c)	$E_{\alpha'}$	b	u	t
	b	1	0	0
	u	0	1	1
	t	0	1	1

d)	$E_{\alpha'}$	b	u	t
	b	1	0	0
	u	0	1	0
	t	0	0	1

In all the cases, $E_{\alpha'}$ is an equivalence relation.

3. $E_{\alpha'}$ is always an equivalence relation, whatever is the domain.

Let D' be an arbitrary set and $I' = (D', \alpha')$.

- **Reflexivity:** $\llbracket \text{Both} \rrbracket_{I'} = 1 \implies \forall x \in D'. E_{\alpha'}(x, x) \leftrightarrow L_{\alpha'}(x, x)$
(by choosing $y = x$).

We also know that, $\llbracket \text{Tot} \rrbracket_{I'} = 1 \implies \forall x \in D'. L_{\alpha'}(x, x)$

Therefore $\forall x \in D'. E_{\alpha'}(x, x)$

- **Symmetry:**

$$\llbracket \text{Both} \rrbracket_{I'} = 1 \implies \forall x \in D'. \forall y \in D'. E_{\alpha'}(x, y) \leftrightarrow (L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x))$$

We also know that for $x, y \in D'$ arbitrary $L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x) \equiv L_{\alpha'}(y, x) \wedge L_{\alpha'}(x, y)$. Therefore we have that for $\forall x \in D'. \forall y \in D'. E_{\alpha'}(x, y) \leftrightarrow E_{\alpha'}(y, x)$ is a tautology.

- **Transitivity:** $\llbracket \text{Both} \rrbracket_{I'} = 1$ implies that for x, y, z arbitrary in D'

$$E_{\alpha'}(x, y) \leftrightarrow (L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x))$$

and

$$E_{\alpha'}(y, z) \leftrightarrow (L_{\alpha'}(y, z) \wedge L_{\alpha'}(z, y))$$

Therefore

$$E_{\alpha'}(x, y) \wedge E_{\alpha'}(y, z) \leftrightarrow (L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, z) \wedge L_{\alpha'}(z, y) \wedge L_{\alpha'}(y, x))$$

which implies

$$E_{\alpha'}(x, y) \wedge E_{\alpha'}(y, z) \rightarrow (L_{\alpha'}(x, z) \wedge L_{\alpha'}(z, x))$$

and finally

$$E_{\alpha'}(x, y) \wedge E_{\alpha'}(y, z) \rightarrow E_{\alpha'}(x, z)$$

4. One of the solution consists in choosing $D_1 = D' / E_{\alpha'}$, the set of equivalence classes of $E_{\alpha'}$, $\alpha_1 = (B_1, T_1, L_1)$ where $B_1 = [b]_{E_{\alpha'}}$, $T_1 = [t]_{E_{\alpha'}}$ and

$$L_1 := \{([x]_{E_{\alpha'}}, [y]_{E_{\alpha'}}) : L_{\alpha'}(x, y) \text{ where } x, y \in D'\}$$

Reflexivity and transitivity are immediate as they have been proven for $L_{\alpha'}$.

Antisymmetry comes from the fact that for arbitrary $[x]_{E_{\alpha'}}, [y]_{E_{\alpha'}} \in D_1$

$$\begin{aligned} & L_1([x]_{E_{\alpha'}}, [y]_{E_{\alpha'}}) \wedge L_1([y]_{E_{\alpha'}}, [x]_{E_{\alpha'}}) \\ \implies & L_{\alpha'}(x, y) \wedge L_{\alpha'}(y, x) \\ \implies & E_{\alpha'}(x, y) \\ \implies & [x]_{E_{\alpha'}} = [y]_{E_{\alpha'}} \end{aligned}$$

Therefore L_1 is a partial order.